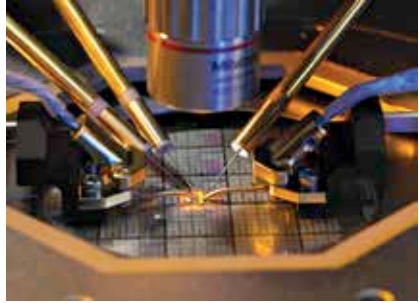


Device test

After finishing the front-end process, the semiconductor devices undergo a multitude of electrical tests which help determine if they properly function. The proportion of the devices on the wafer that are found to function properly is called the yield. Usually, the manufacturers are secretive about the yields. However, it's said that the yield could be as low as 30%. One of the many reasons for a low yield is process variation.

The chips on the wafer are usually tested using an electronic presser. The presser has tiny probes that would be pressed against the chip. Each bad chip that's identified would be marked with a dye drop. Electronic dye marking is done if the wafer test data gets logged into a central computer database and the chips get sorted into virtual bins pertaining to predefined test limits. The binning data thus obtained could be logged or graphed on a wafer map. This helps trace any manufacturing defects and also to mark the faulty chips. The map could also be used for wafer assembly and packaging.



Testing it, one wafer at a time!

After packaging, the chips are tested yet again. This is because there could be anomalies like missing bond wires or the analog performance getting altered by the package. This test is indeed the final test.

Testing times in a fab could vary from just a few milliseconds to a couple of seconds. The test software is usually optimized to reduce the testing time. Multiple chip or multi-site testing is also a possibility—many testers possess the resources to run nearly all the tests in parallel.

To facilitate speedier testing and bring down testing costs, chips are usually



Power on, test!